



Basic Arithmetic Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Simplify:

a. $\frac{x+2y}{3} - \frac{2x-y}{4}$

$$(a) \frac{x+2y}{3} - \frac{2x-y}{4} = \frac{4(x+2y) - 3(2x-y)}{12} = \frac{4x+8y-6x+3y}{12} = \frac{11y-2x}{12}$$

2

b. $\frac{(2x^{-4})^3}{5x^{-1}}$

$$(b) \frac{(2x^{-4})^3}{5x^{-1}} = (8x^{-12}) \div \frac{5}{x} = \frac{8}{x^{12}} \times \frac{x}{5} = \frac{8}{5x^{11}}$$

2

2 Simplify $\frac{1}{\sqrt{3}} + \frac{1}{3+\sqrt{3}}$

$$\begin{aligned} \frac{1}{\sqrt{3}} + \frac{1}{3+\sqrt{3}} &= \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{1}{3+\sqrt{3}} \times \frac{3-\sqrt{3}}{3-\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} + \frac{3-\sqrt{3}}{6} \\ &= \frac{2\sqrt{3}+3-\sqrt{3}}{6} = \frac{\sqrt{3}+3}{6} \end{aligned}$$

2

3 Simplify the expression: $\frac{6x^3 \times 5x^{-1}}{2x^3 \times 5x^{-2}}$

$$\frac{6x^3 \times 5x^{-1}}{2x^3 \times 5x^{-2}} = \frac{30x^2}{10x^1} = 3x^1 = 3x$$

1

4 Express the following as a single, simplified, surd expression: $\sqrt{3}^3 + \sqrt{12}$

$$\sqrt{3}^3 + \sqrt{12} = \sqrt{27} + \sqrt{12} = \sqrt{9} \times \sqrt{3} + \sqrt{4} \times \sqrt{3} = 3\sqrt{3} + 2\sqrt{3} = 5\sqrt{3}$$

1

5 Express the following as a single, simplified algebraic fraction: $\frac{a+2}{a^2+5a} - \frac{5a}{a^2-25}$

$$\begin{aligned} \frac{a+2}{a^2+5a} - \frac{5a}{a^2-25} &= \frac{a+2}{a(a+5)} - \frac{5a}{(a+5)(a-5)} \\ &= \frac{(a+2)(a-5)}{a(a-5)(a+5)} - \frac{5a^2}{a(a+5)(a-5)} \\ &= \frac{a^2-3a-10-5a^2}{a(a-5)(a+5)} \\ &= \frac{-3a-10-4a^2}{a(a-5)(a+5)} = \frac{-3a-10-4a^2}{a^3-25a} \end{aligned}$$

2

6 Expand and simplify $(1 + \sqrt{3})(4 - 2\sqrt{3})$

$$4 - 2\sqrt{3} + 4\sqrt{3} - 2\sqrt{3}\sqrt{3} = 2\sqrt{3} - 2$$

1

7 Simplify $\frac{x}{3} - \frac{2x+1}{x}$

$$\frac{x^2 - 3(2x+1)}{3x} = \frac{x^2 - 6x - 3}{3x}$$

1

8 Simplify $(7 + \sqrt{3})(7 - \sqrt{3})$

$$(7 + \sqrt{3})(7 - \sqrt{3}) = 7^2 - (\sqrt{3})^2 = 49 - 3 = 46$$

1

9 Simplify $\frac{1}{1+\sqrt{5}}$

$$\frac{1}{1+\sqrt{5}} \times \frac{1-\sqrt{5}}{1-\sqrt{5}} = \frac{1-\sqrt{5}}{1^2 - (\sqrt{5})^2} = \frac{1-\sqrt{5}}{-4}$$

1

- 10 For x and y rational numbers and $(\sqrt{x} + \sqrt{y})^2 = x + y + 2\sqrt{xy}$, write $\sqrt{16 + 2\sqrt{55}}$ in the form 2

$$\sqrt{x} + \sqrt{y}$$

$$\begin{aligned} (\sqrt{x} + \sqrt{y})^2 &= x + y + 2\sqrt{xy} \\ \therefore \sqrt{x} + \sqrt{y} &= \sqrt{x+y+2\sqrt{xy}} \\ \therefore x+y &= 16 \\ + \quad xy &= 55 \\ \therefore xy &= 11, 5 \\ \therefore \sqrt{16+2\sqrt{55}} &= \sqrt{11} + \sqrt{5} \end{aligned}$$

- 11 Find integers a and b such that $\frac{1}{\sqrt{5}-2} = a + b\sqrt{5}$. 2

$$\begin{aligned} \frac{1}{\sqrt{5}-2} &= a+b\sqrt{5} \\ \frac{1}{\sqrt{5}-2} \times \frac{\sqrt{5}+2}{\sqrt{5}+2} &= a+b\sqrt{5} \\ \frac{\sqrt{5}+2}{(\sqrt{5})^2-(2)^2} &= a+b\sqrt{5} \\ \frac{\sqrt{5}+2}{5-4} &= a+b\sqrt{5} \end{aligned} \quad \left| \begin{array}{l} \sqrt{5}+2 = a+b\sqrt{5} \\ \text{Rearranging and} \\ \text{Comparing} \\ a=2 \text{ and } b=1 \end{array} \right.$$

- 12 A mixture is to be made of two foods A and B . The two foods

Ounces per pound of nutrient

A, B contain nutrients P and Q as shown below:

The mixture contains 8 ounces of P and 6 ounces of Q .

For example: one part of A and 3 parts of B will make 8 ounces of P

i.e. $A + 3B = 8$

Food	P	Q
A	1	2
B	3	1

- a. Use a similar method to form an equation for nutrient Q . 1
- b. Solve the two equations simultaneously to find the values of A and B . 3

$$\begin{aligned} A+3B &= 8 \quad \text{--- (1) } \times 2 & \text{Sub } B=2 \text{ in (2)} \\ 2A+B &= 6 \quad \text{--- (2)} & 2A+2 &= 6 \\ 2A+6B &= 16 & \therefore 2A &= 4 \\ (-) 2A+B &= 6 & A &= 2 \\ \hline 5B &= 10 & \therefore A=2 \text{ and } B=2 \\ \hline B &= 2 \end{aligned}$$

- 13 Solve $2x^2 - 5x + 2 \geq 0$. 2

$$\begin{aligned} (2x-1)(x-2) &\geq 0 \\ x &\leq \frac{1}{2} \quad \text{or} \quad x \geq 2 \end{aligned}$$

14

- a. Solve the equation $x^2 - 6 = x$.

1

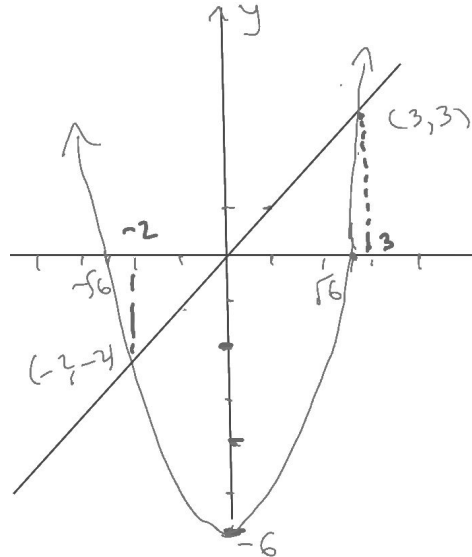
$$x^2 - x - 6 = 0$$

$$(x - 3)(x + 2) = 0$$

$$x = 3 \text{ or } -2$$

- b. Sketch the graphs of $y = x^2 - 6$ and $y = x$. Label the coordinates of any x and y intercepts and any points of intersection of the graphs.

3



- c. Find the values of x for which $x^2 - 6 > x$.

2

$$x < -2 \text{ or } x > 3$$



Basic Arithmetic Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Simplify:

a. $\frac{4a}{7} \times \frac{21}{5a^2}$ $\frac{4\cancel{a} \times 21^{\cancel{3}}}{7 \times 5a^2} = \frac{12}{5a}$

1

b. $\frac{5}{x^2-1} \div \frac{2}{x-1}$

(ii) $\frac{5}{x^2-1} \div \frac{2}{x-1}$
 $= \frac{5}{(x+1)(x-1)} \times \frac{(x-1)}{2}$
 $= \frac{5}{2(x+1)}$ or $\frac{5}{2x+2}$

1

c. $\frac{3}{b+2} \times \frac{b^2+2b}{6a-3}$

(iii) $\frac{3}{b+2} \times \frac{b^2+2b}{6a-3}$
 $= \frac{3}{b+2} \times \frac{b(b+2)}{3(2a-1)}$
 $= \frac{b}{2a-1}$

d. $\frac{3ab^2}{5xy} \div \frac{12ab-6a}{x^2y+2xy^2}$

$\frac{3ab^2}{5xy} \div \frac{12ab-6a}{x^2y+2xy^2}$
 $= \frac{3ab^2}{5xy} \times \frac{xy(x+2y)}{2a(2b-1)}$
 $= \frac{b^2(x+2y)}{10(2b-1)}$
 $= \frac{b^2x+2yb^2}{20b-10}$

1

2

2 Simplify $\frac{x}{x^2-4} - \frac{4}{x-2}$

$= \frac{x}{(x-2)(x+2)} - \frac{4}{x-2}$
 $= \frac{x - 4(x+2)}{(x-2)(x+2)}$
 $= \frac{x - 4x - 8}{(x-2)(x+2)}$
 $= \frac{-3x-8}{(x-2)(x+2)}$

2

3 Simplify:

a. $\frac{a+4}{3} - \frac{a}{4}$

$\frac{a+4}{3} - \frac{a}{4}$
 $= \frac{4(a+4) - 3a}{12}$
 $= \frac{4a+16-3a}{12} = \frac{a+16}{12}$

1

b. $\frac{1}{a^2+2a+1} + \frac{1}{a+1}$

$$\begin{aligned} & \text{(ii)} \quad \frac{1}{a^2+2a+1} + \frac{1}{a+1} \\ &= \frac{1}{(a+1)^2} + \frac{1}{a+1} \\ &= \frac{1 + (a+1)}{(a+1)^2} \\ &= \frac{a+2}{(a+1)^2} \end{aligned}$$

c. $x - \frac{x^2}{x+2}$

$$\begin{aligned} \frac{x - x^2}{x+2} &= \frac{x(x+2) - x^2}{x+2} \\ &= \frac{x^2 + 2x - x^2}{x+2} \\ &= \frac{2x}{x+2} \end{aligned}$$

d. $\frac{1}{x-1} - \frac{2}{x}$

$$\begin{aligned} \frac{1}{x-1} - \frac{2}{x} &= \frac{x - 2(x-1)}{x(x-1)} \\ &= \frac{x - 2x + 2}{x(x-1)} \\ &= \frac{2-x}{x(x-1)} \end{aligned}$$

4 Write 1.627 as a rational number.

$$\begin{aligned} x &= 1.627 \\ x &= 1.62727... \\ 10x &= 16.2727... \\ 1000x &= 1627.2727... \\ 1000x - 10x &= 1627.2727... - 16.2727... \\ 990x &= 1627 - 16 \\ 990x &= 1611 \\ x &= \frac{1611}{990} \\ x &= 1 \frac{61}{110} \end{aligned}$$

5 DA Tuition wants to sell a new sports shirt to its students at \$40 each. To manufacture the shirts, the centre pays a fixed cost of \$595, as well as \$33 per shirt. By writing a pair of simultaneous equations, calculate how many shirts the centre must sell in order to break-even. Let x be the number of shirts.

$$\text{Total cost for the school} = \$595 + 33x$$

$$595 + 33x = 40x$$

$$7x = 595$$

$$\therefore x = 85 \text{ shirts}$$

6 Simplify: $\frac{2p^2-3p-2}{8p^3+1}$

$$\begin{aligned} & \frac{2p^2-3p-2}{8p^3+1} \\ &= \frac{(2p+1)(p-2)}{(2p+1)(4p^2-2p+1)} \\ &= \frac{p-2}{4p^2-2p+1} \end{aligned}$$

7 Simplify: $(\sqrt{3} - 1)(2\sqrt{3} + 5)$

1

$$\begin{aligned} & \sqrt{3}(2\sqrt{3} + 5) - 1(2\sqrt{3} + 5) \\ &= 6 + 5\sqrt{3} - 2\sqrt{3} - 5 \\ &= 1 + 3\sqrt{3} \end{aligned}$$

8 Find the values of a and b (in simplified form) such that $\frac{3}{4-\sqrt{2}} = a + \sqrt{b}$.

2

$$\begin{aligned} & \frac{3}{4-\sqrt{2}} \times \frac{4+\sqrt{2}}{4+\sqrt{2}} \\ &= \frac{12+3\sqrt{2}}{16-4} \\ &= \frac{12+3\sqrt{2}}{14} \\ &= \frac{6}{7} + \frac{3}{14}\sqrt{2} \end{aligned}$$

$$\begin{aligned} & \therefore a = \frac{6}{7}, \sqrt{b} = \frac{3}{14}\sqrt{2} \\ & b = \left(\frac{3}{14}\sqrt{2}\right)^2 \\ &= \frac{9 \times 2}{196} \\ &= \frac{18}{196} \\ & \therefore b = \frac{9}{98} \end{aligned}$$

9 Simplify:

a. $\frac{m^2+2m-8}{m^2-4} = \frac{(m-2)(m+4)}{(m-2)(m+2)} = \frac{m+4}{m+2}$

1

b. $\frac{1-t^2}{t-1} = \frac{(1+t)(1-t)}{-(1-t)} = -(1+t)$

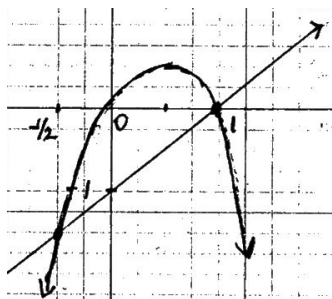
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c. $\frac{x^3+8}{x^2-2x+4} = \frac{(x+2)(x^2-2x+4)}{(x^2-2x+4)}$

2

10 a. Sketch the graph of $y = 2x - 2x^2$ and $y = x - 1$ on the same number plane.

2



b. Using your graph from part (i), or otherwise solve the inequality $x - 1 < 2x - 2x^2$.

2

$$\begin{aligned} & \text{pts. of intersection} \\ & x-1 = 2x-2x^2 \\ & 2x^2 - x - 1 = 0 \\ & (2x+1)(x-1) = 0 \\ & x = -\frac{1}{2} \quad x = 1 \\ & \text{answer graphically (line below parabola)} \\ & -\frac{1}{2} < x < 1 \end{aligned}$$